Review of "Selection Functionals on Path Space for Stochastic Processes Without a Generator: Finite-Grid Gibbs Selection and Self-Normalized Estimation"

Overall Assessment

This manuscript develops a finite-grid path-space framework for stochastic systems whose observed two-time transition families violate the Chapman-Kolmogorov relation. Its central construction uses a discretized Onsager-Machlup action, evaluated against a deliberately imperfect surrogate model, to reweight paths generated by an external proposal. The resulting measure is a Gibbs reweighting of the proposal, and expectations under this selected measure are estimated using self-normalized importance sampling. The manuscript establishes consistency, asymptotic normality, finite-sample concentration bounds, variance and effective-sample-size diagnostics, and a conditional comparison with the law of a target process. It also includes a detailed Ornstein-Uhlenbeck example in which the principal quantities are available in closed form.

The manuscript is generally clearly structured and is considerably more careful than its broad motivation might initially suggest. In particular, it correctly distinguishes the selected measure from the law of the true process, restricts its rigorous construction to a fixed finite grid, and explicitly acknowledges that the relation between the selected measure and the target law is not established by the estimator theory. The self-normalized importance-sampling results are largely standard but appear mathematically sound, and the worked Gaussian example provides a useful check on the definitions and asymptotic diagnostics. The explicit discussion of weight degeneracy and the distinction between asymptotic and finite-sample claims are also strengths.

The principal difficulty is that the connection between the proposed estimator and indivisible stochastic dynamics remains largely motivational rather than mathematical. Theorems A-C depend only on the proposal measure and the surrogate weight; neither the target transition family nor its Chapman-Kolmogorov deficit enters their statements or proofs. The estimator therefore consistently evaluates expectations under the constructed selected measure, but it does not unconditionally estimate expectations under the target process. Theorem D supplies such a connection only by directly assuming that the selected and true grid laws are close in total variation. Consequently, the main estimator theory is an application of established self-normalized importance-sampling arguments to a particular Gibbs weight, while the genuinely process-specific approximation problem is left open.

There is also a significant logical issue in the interpretation of Proposition 7.3. The proposition gives an operator-norm obstruction to uniformly approximating an indivisible transition family by a divisible transition family. It does not imply that the selected measure $Q_{\nu,M}$ cannot coincide with, or closely approximate, the true grid law, because $Q_{\nu,M}$ depends jointly on the surrogate weight and the external proposal and is not itself the path law generated by the surrogate. The manuscript in fact later acknowledges that Proposition 7.3 gives no quantitative information about the total-variation distance appearing in hypothesis (D1). Statements that the kernel obstruction forces a nonzero selection bias, or that no choice of divisible surrogate can drive that bias to zero, therefore do not follow and should be removed or substantially reformulated.

Overall Conclusion

Further clarification is needed concerning the novelty of the statistical results, the calibration proposal when the diffusion coefficient is treated as an estimable parameter, the interpretation of the continuous-path degeneracy, and several practical claims concerning effective sample size and sample-size thresholds. These issues do not invalidate the finite-grid estimator or its principal self-normalized importance-sampling results, but they materially affect the manuscript's interpretation and the strength of its claimed contribution. The paper could provide a coherent methodological account of finite-grid Gibbs selection following substantial revision, particularly if its claims are confined to the selected measure, the obstruction result is separated from the unproved truth-link problem, and the statistical contribution is positioned more precisely relative to existing importance-sampling theory.

Major Comments

The opening material is generally clear and already contains several sensible qualifications. The distinction between the absence of a generator on the observed state space and the possible existence of hidden-variable realizations or a path-space law is particularly important and is handled much better here than in the earlier papers. However, several claims about the consequences of indivisibility and the novelty of the construction remain too strong.

The paper does not yet establish uncertainty quantification for the true non-Markovian system.

The significance statement concludes that the framework "supplies uncertainty quantification" for non-Markovian systems. What is established, however, is uncertainty quantification for expectations under the selected measure $Q_{\nu,M}$, which is constructed from the proposal and surrogate action. Its relationship to the true process is assumed later through condition (D1), rather than derived.

The opening claims should therefore consistently distinguish

$$\text{uncertainty under } Q_{\nu,M}$$

from

$$\text{uncertainty concerning } P_{\text{true},M}.$$

Until a quantitative truth-link is established, the former does not automatically provide the latter.

The main estimator does not use the indivisible target family.

The introduction states that the framework applies when one has an empirical transition family that violates Chapman-Kolmogorov and an external candidate-path mechanism. However, the estimator itself requires only the proposal Π_M, the surrogate action, and the observable. The transition family $\Gamma(t \leftarrow s)$ and its deficit do not enter the construction.

This does not invalidate the estimator, but it means that indivisibility presently supplies the motivation rather than an essential mathematical input. The manuscript should make this structural separation explicit and explain what operational role the empirical transition family plays in selecting or validating the proposal, surrogate, or temperature.

The claimed distinction from generator-based Bayesian methods is overstated.

The absence of a generator reproducing the target transition family does not imply that generator-based likelihoods or posteriors cannot be constructed on the observed state space. A deliberately misspecified Markov model can still define a likelihood and posterior; what fails is the claim that such a model correctly reproduces the target's two-time dynamics.

Accordingly, statements such as

> "$w_\nu \, d\Pi_M$ is not the unnormalized posterior of any generator-based Bayesian model"

are too strong. The defensible conclusion is that it is not derived from a correctly specified generator-based model reproducing the given transition family. This distinction should be maintained in the significance statement, introduction, and comparison with previous SNIS theory.

The obstruction result is presented as more closely connected to the truth-link than it actually is.

The abstract and contribution summary present the conditional total-variation transfer and the kernel-level obstruction as complementary parts of one result. However, the obstruction concerns operator-norm distance between two-time transition families, whereas hypothesis (D1) concerns total-variation distance between full grid-path measures. No implication between these quantities is established.

The opening summary should not suggest that the obstruction explains or controls the selection bias. At most, it shows that a divisible transition family cannot uniformly reproduce the target kernels below a deficit-dependent threshold. Whether this has any consequence for

$$d_{\mathrm{TV}}(Q_{\nu,M}, P_{\mathrm{true},M})$$

remains open.

The novelty relative to ordinary SNIS should be stated more modestly.

The consistency, CLT, concentration, and effective-sample-size results are standard consequences of applying self-normalized importance sampling to the density ratio

$$\rho_\nu = \frac{w_\nu}{Z_\nu}.$$

The two distinctions listed in the opening discussion concern the interpretation and motivation of the weight, not a new form of SNIS theory. The manuscript's original contribution therefore appears to lie principally in proposing the finite-grid OM-inspired selection weight and integrating it with the indivisibility programme, rather than in the general estimator theory itself.

The contribution summary should distinguish new results from standard SNIS results specialized or restated for this construction.

"Quadratic functional" is inaccurate for nonlinear surrogates.

The introduction states that a parameterized drift-diffusion surrogate "defines a quadratic functional on each candidate path." The mismatch is squared, but if $b_\nu(\xi_m)$ is nonlinear, the resulting action is not generally quadratic in the path variables. The phrase should be replaced by something such as "a functional containing a quadratic increment-mismatch term."

The scope of Proposition 2.1 should be carried into its conclusion.

Proposition 2.1 correctly shows that no Itô diffusion on the observed state space can reproduce an indivisible two-time family. The conclusion that neither the tube-density nor Freidlin-Wentzell role is "available for the target" is nevertheless broader than the proof.

A non-Markovian target with a well-defined path law may possess its own small-tube asymptotics or large-deviation rate functional. What is unavailable is the classical drift-diffusion form derived from a pair (b, σ) reproducing the target kernels. The proposition should state this narrower conclusion explicitly.

Equation (4) requires a precise definition of the reference path.

The tube-probability ratio is written relative to a path ξ°, but ξ° is not defined. Its regularity, initial condition, and role as the reference path should be stated, together with the precise tube norm and assumptions under which the limit holds. The current text makes a theorem from the OM literature appear more unconditional than it is.

The explanation of the divergence term is too compressed.

The statement that $\nabla \cdot b$ is simply "the Itô correction from the Girsanov density" is imprecise. Girsanov provides the change-of-measure density, while the divergence contribution arises when converting this representation into the Onsager-Machlup small-tube functional. This sentence should either be explained more carefully or stated less specifically.

The opening material is repetitive.

The abstract, significance statement, introduction, and seven-item contribution list repeat the same selected-measure/truth-link qualifications at considerable length. Some repetition is useful because the distinction is central, but the opening could be shortened substantially without losing mathematical content.

This part of the manuscript is mathematically stronger than the opening discussion. Proposition 2.3 is correct under the definition adopted by the author, and the consistency, central-limit, and concentration arguments are essentially standard but sound. The principal issues concern the interpretation of the selected measure and whether the stated sequential and Bayesian connections actually follow from the assumptions.

The generalized-Bayes interpretation is not formulated correctly.

Remark 3.8 describes $S_{\nu,M}$ as a loss "acting on a single observation (the candidate path itself)." However, the candidate path is the variable being distributed under $Q_{\nu,M}$, not external observed data conditioning that distribution. Therefore,

$$dQ_{\nu,M}(\xi) \propto e^{-\lambda S_{\nu,M}[\xi]}\, d\Pi_M(\xi)$$

certainly defines a Gibbs change of measure, but it is not automatically a generalized-Bayes posterior in the ordinary inferential sense.

The analogy could be retained at the level of mathematical form, but the manuscript should distinguish a Gibbs measure on candidate paths from a posterior update based on observed data. If observed data are intended to enter through the calibration of ν, that is a separate operation introduced only later in Section 8.4.

The claim that the selected measure cannot come from any generator-based model is too strong.

Following Definition 3.7, the manuscript states that under indivisibility the measure $w_\nu\, d\Pi_M$ is not the path-space law of "any generator-based model on the observed description." This does not follow from the indivisibility of the target. That property only prevents a generator-based model from reproducing the specified target kernels.

The selected measure depends on both w_ν and Π_M, and it may itself possess Markovian or generator-based structure. Indeed, for suitable Gaussian proposals and quadratic weights, the selected finite-grid measure may be Markov even though the motivating target is not. The statement should therefore be weakened to say that $w_\nu\, d\Pi_M$ is not established as the law of a generator-based model reproducing the target transition family.

Local factorization of the weight does not by itself justify a particle-filter implementation.

Remark 3.9 correctly observes that the discrete action is additive over increments and that w_ν factorizes into local terms. However, the proposal Π_M has been left completely abstract and is assumed only to admit independent sampling of full paths. Unless Π_M also has an appropriate sequential factorization, the selected measure cannot necessarily be sampled or updated through an ordinary particle filter merely because the weight factorizes.

The manuscript should either impose a sequential structure on the proposal or state more cautiously that the factorization may be useful for sequential implementations when the proposal admits compatible conditional sampling.

The terminology "Euler-Maruyama discretization" is potentially misleading.

Equation (7) is a left-point discretization of the continuous Onsager-Machlup functional. It is not the negative log-density of the Euler-Maruyama transition scheme: the latter contains Gaussian normalization terms but not the displayed divergence correction in this form.

Because the manuscript expressly denies that $S_{\nu,M}$ is a likelihood, the definition itself is legitimate. Nevertheless, it would be more accurate to call (7) a left-point finite-grid discretization of the OM functional rather than an Euler-Maruyama discretization, or else explain precisely which convention is intended.

Proposition 2.3 is correct, but its interpretation should remain tied to the chosen pointwise extension.

The proof that Brownian, Brownian-bridge, OU, and suitable diffusion proposals assign zero probability to absolutely continuous paths is sound. It follows that the weight defined in (6), with $S^{\mathrm{OM}} = +\infty$ away from absolutely continuous paths, vanishes almost surely.

This demonstrates the degeneracy of that particular attempt to interpret the classical OM functional as an ordinary pointwise importance weight. It should not be presented as a degeneracy of continuous-path OM theory itself, whose classical role is based on tube probabilities or Cameron-Martin comparisons rather than pointwise evaluation on typical diffusion paths.

The bound in Lemma 3.6 can be sharpened.

From $0 < w_\nu \le W_{\max}$, one has

$$w_\nu^2 \le W_{\max} w_\nu.$$

Consequently,

$$\chi^2(Q_{\nu,M} \,\|\, \Pi_M) = \frac{\mathbb{E}_{\Pi_M}[w_\nu^2]}{Z_\nu^2} - 1 \le \frac{W_{\max}}{Z_\nu} - 1.$$

This is sharper than the stated bound

$$\chi^2(Q_{\nu,M} \,\|\, \Pi_M) \le \frac{W_{\max}^2}{Z_\nu^2} - 1.$$

The current inequality is valid, but the stronger bound follows immediately from the same hypothesis and would give a more meaningful diagnostic when $Z_\nu \ll W_{\max}$.

Theorem 4.1 is sound, but "minimal moment conditions" should be softened.

The proof of strong consistency is standard and correct. The CLT assumptions are also sufficient, but they are not stated in their weakest form: it would be enough to assume integrability of the denominator and finite variance of

$$(F - F^*)w_\nu,$$

rather than separately imposing all of the displayed second-moment conditions. The phrase "minimal moment conditions" in the contribution summary should therefore be replaced by "standard moment conditions" or "mild sufficient moment conditions."

The asymptotic MSE result is mathematically valid but not independently operational.

Part (iii) of Theorem 4.1 correctly uses uniform integrability to upgrade distributional convergence to convergence of second moments. However, uniform integrability of

$$N\big(\widehat{F}_N - F^*\big)^2$$

is itself a substantial additional assumption, and no directly verifiable conditions implying it are provided. This should be acknowledged. Otherwise, the theorem may appear to derive an asymptotic MSE result from the second-moment conditions in part (ii), which it does not.

Theorem 5.1 appears correct, although the bound is conservative.

The decomposition into numerator and denominator events, use of Chebyshev and Cantelli inequalities, and bounded-observable MSE estimate are correct. The resulting bounds can exceed the trivial probability or squared-error bounds for small N, so they should be understood as asymptotic-order estimates rather than uniformly informative finite-sample guarantees. A brief comment to this effect would improve their practical interpretation.

Assumption (P2) need not be part of the general proposal conditions.

The computable-density assumption is stated as a proposal condition even though the estimator needs only samples and the manuscript says the assumption is used solely in the worked example. It would be cleaner to remove (P2) from the standing framework and introduce it locally in Section 9.

The temperature parameter requires some interpretive clarification.

Since λ controls both concentration of the selected measure and weight degeneracy, the choice of λ materially determines the estimand $F^*(\nu)$. At present it is simply fixed without a selection principle. The manuscript should emphasize that different temperatures define different target measures, rather than merely changing the numerical efficiency of estimating one fixed quantity.

This is the most important part of the manuscript. Theorem 6.1, Theorem 7.1, and Proposition 7.3 are individually largely correct, and the numerical quantities in the OU example can be reproduced from the stated precision matrices. However, the manuscript draws a conclusion from Proposition 7.3 that the proposition does not support. The calibration section also contains a significant structural problem when the diffusion coefficient is estimated.

The kernel obstruction does not imply a nonzero selection bias.

Section 7.2 opens with the claim that

> "no choice of divisible surrogate drives the selection bias to zero, because divisible families are uniformly bounded away from an indivisible target's kernels."

This does not follow from Proposition 7.3. The proposition compares the target transition family Γ with a divisible surrogate transition family $\widetilde{\Gamma}$. The selection bias in Theorem 7.1 instead compares the true grid-path law $P_{\mathrm{true},M}$ with

$$dQ_{\nu,M} = \frac{w_\nu}{Z_\nu}\, d\Pi_M.$$

The measure $Q_{\nu,M}$ is determined jointly by the weight and the proposal; it is not the path law generated by $\widetilde{\Gamma}$. It can therefore coincide with $P_{\mathrm{true},M}$ even when the surrogate kernels remain separated from Γ.

Indeed, for a positive weight satisfying the required integrability conditions, one could formally choose a proposal proportional to $w_\nu^{-1} P_{\mathrm{true},M}$, in which case the Gibbs reweighting returns $Q_{\nu,M} = P_{\mathrm{true},M}$. This may not be a practical construction, but it disproves the claimed logical implication.

The manuscript itself subsequently states that Proposition 7.3 "says nothing quantitative" about the total-variation distance in (D1). That acknowledgement directly conflicts with the opening sentence of Section 7.2. The obstruction should be presented as a separate kernel-level result, not as an explanation for unavoidable selection bias or for the conditional structure of Theorem 7.1.

The proposed "open bridge" cannot depend only on an invariant of the target without restrictions on the proposal and surrogate.

Remark 7.2 asks whether the tolerance $\eta_{\nu,M}$ can be controlled by the integrated deficit or memory rank of the target. No bound depending only on an invariant of the target can control

$$d_{\mathrm{TV}}(Q_{\nu,M}, P_{\mathrm{true},M})$$

for arbitrary proposal-surrogate pairs, because a poor proposal or extreme temperature can make $Q_{\nu,M}$ arbitrarily far from the true law even when the target is divisible.

The remark includes the phrase "for principled choices of proposal and surrogate," but this class is not defined. A meaningful open problem requires an explicit construction or admissible class of proposal-surrogate pairs and a bound containing terms that measure proposal and surrogate error separately. Otherwise, the conjectured bridge is not yet a mathematically definite problem.

The explanation of why calibration is not a KL projection is incorrect.

Section 8.4 states that ν^* cannot be interpreted as a Kullback-Leibler projection because "no true generator-based law exists." A KL projection does not require the true law itself to be generator-based. It requires only a true probability measure $P_{\mathrm{true},M}$, a model family $P_{\nu,M}$, and the relevant absolute-continuity and integrability conditions.

The manuscript has already assumed the existence of $P_{\mathrm{true},M}$ in Section 7. If $S_{\nu,M}$ were the negative log-density of a normalized surrogate path model, minimizing its expectation could potentially be interpreted as a misspecified-model KL projection even though the truth is non-Markovian. The genuine obstacle is that the proposed action is not established as such a normalized negative log-density, not that the true process lacks a generator.

Calibration of the diffusion coefficient using (21) is generally ill-posed.

The parameter ν includes σ_ν, but the empirical criterion in (21) contains the quadratic residual term without the normalization term associated with a Gaussian increment density. For example, under a scalar rescaling $\sigma_\nu = cI$, the kinetic part decreases like c^{-2}, whereas the divergence term is unaffected. Minimizing $S_{\nu,M}$ can therefore favour arbitrarily large diffusion scales or push c to the upper boundary of the compact parameter set.

The generic assumptions of identification and compactness listed afterward do not explain why this criterion identifies a meaningful diffusion coefficient. The author should therefore do one of the following:

- treat σ_ν as fixed during calibration;
- add and justify the appropriate normalization term; or
- provide conditions showing that the proposed objective has an interior, identifiable population minimizer.

The Mori-Zwanzig subsection does not yet characterize a surrogate family.

Section 8.3 provides a qualitative description of truncating the memory integral and dropping the fluctuating force, but it does not derive an explicit drift-diffusion pair $(b_{\nu,P}, \sigma_{\nu,P})$. In particular, simply dropping the fluctuating force does not automatically leave a nondegenerate diffusion coefficient of the kind required by Definition 3.3.

This subsection should either provide an explicit closure and verify that it satisfies the surrogate assumptions, or be presented as a possible source of surrogate models rather than as one of three characterized surrogate families.

Theorem 6.1 is mathematically correct.

The identities

$$\sigma^2(\nu) = \mathbb{E}_{Q_{\nu,M}}\left[(F - F^*)^2 \rho_\nu\right], \qquad \mathbb{E}_{Q_{\nu,M}}[\rho_\nu] = 1 + \chi^2$$

follow directly from the change of measure, and the bounded-observable estimate is valid. The effective-sample-size limit and the divergence ordering

$$e^{D_{\mathrm{KL}}(Q_{\nu,M}\|\Pi_M)} \le 1 + \chi^2(Q_{\nu,M}\|\Pi_M)$$

are also correct.

Calling $1 + \chi^2$ a "variance-inflation factor" should nevertheless remain qualified: it is an observable-independent upper-bound factor, not the actual inflation for a particular F, as the manuscript correctly notes.

The practical ESS thresholds in Remark 6.2 are insufficiently supported.

The claims that the estimator is generally unreliable when $\widehat{N}_{\mathrm{eff}} \lesssim 10^2$, and that Pareto-smoothed importance sampling is recommended whenever $\widehat{N}_{\mathrm{eff}}/N \lesssim 10^{-2}$, are presented as universal operational rules. Their validity depends on the observable, weight-tail behaviour, and required accuracy.

Moreover, Pareto-smoothed importance sampling is normally assessed using tail-shape diagnostics, and modifying the weights changes the estimator to which Theorems A-C apply. These statements should be cited and qualified or replaced with a more cautious warning that small estimated ESS indicates possible weight degeneracy.

Theorem 7.1 is correct but contains no process-specific estimate.

The bound

$$|F^*(\nu) - F_{\mathrm{true}}| \le 2F_{\max}\eta_{\nu,M}$$

is the standard total-variation estimate and is proved correctly. However, all information concerning the true process is contained in the assumed tolerance $\eta_{\nu,M}$. Describing the result as a "truth link" is acceptable only if the text makes clear that it transfers an independently established approximation bound rather than deriving one.

The requirement that $\eta_{\nu,M}$ be "known" is particularly strong. The manuscript should explain how this quantity might be established or estimated in an intended application.

Proposition 7.3 is correct under its contraction assumptions.

The three-term estimate follows by inserting the divisible factorization and applying the triangle inequality together with the relevant contraction bounds. The resulting $D/3$ worst-pair lower bound is valid.

However, a diffusion semigroup is not automatically contractive on $L^2(\Omega, \mu)$ for an arbitrary reference measure μ. The manuscript does qualify the example by requiring that the surrogate semigroup be a contraction, but it should state conditions on μ, the generator, and any boundary behaviour under which this property holds.

The interpretation of the MZ obstruction is again too broad.

Proposition 7.3 shows that a fixed indivisible family with positive deficit cannot be uniformly approximated arbitrarily closely by divisible families in the stated operator norm. It does not show that a Mori-Zwanzig closure cannot be a useful approximation in another metric, for particular observables, over restricted time pairs, or in a parameter regime where the target deficit tends to zero.

The statement that the MZ surrogate is "not an approximation scheme that converges to the truth" should therefore be limited to uniform approximation of the fixed two-time family in the operator norm used in Proposition 7.3.

The closed-form OU calculations appear correct.

For the stated parameters, the precision-matrix calculation gives approximately

$$(\Sigma_\lambda)_{MM} = 0.158061, \qquad F^*(\nu) = -0.841939,$$

and the reported values

$$\chi^2 \approx 1372.4, \qquad D_{\mathrm{KL}} \approx 4.860, \qquad Z_\nu \approx 7.87 \times 10^{-8}, \qquad \sigma^2(\nu) \approx 30.89$$

are mutually consistent. This is a genuine strength of the paper.

The worked example illustrates the selected measure rather than the indivisible target problem.

No indivisible target family is specified in the example, and the example tests only the numerical behaviour of an SNIS estimator for a Gaussian-selected measure. It therefore validates the estimator calculations but does not demonstrate that the method recovers observables of an indivisible stochastic process.

The manuscript should describe it as an exact benchmark for the finite-grid selection estimator, rather than as validation of the proposed truth link or of the method's usefulness for indivisible dynamics.

The numerical calculations in the worked example appear correct, including the grid-refinement values. The main problems in this final block arise in Section 10: the claimed distinctions from misspecified-model theory and path-space variational inference rely on conflating "absence of a generator" with "absence of a true probability law." The paper itself has already explained that those are different statements.

The comparison with misspecified-model theory is based on a false distinction.

Section 10.1 states that misspecified-model theory assumes a well-defined true law P_0, whereas in the present setting "there is no generator-based P_0 to score against." However, misspecified-model theory does not require the true law P_0 to belong to a generator-based model family. Its purpose is precisely to study estimation when P_0 lies outside the fitted family.

The manuscript itself assumes a true grid law $P_{\mathrm{true},M}$ in Section 7. Once that law exists, a KL projection onto a surrogate family may be defined whenever the required absolute-continuity and integrability conditions hold. The fact that $P_{\mathrm{true},M}$ is not generated by a Markov diffusion on the observed state space does not prevent it from serving as the true law in misspecified-model theory.

The distinction should therefore be reformulated. The genuine difference is that the proposed objective $S_{\nu,M}$ is not shown to be the negative log-density of a normalized surrogate path family, not that the truth lacks a generator.

Path-space relative entropy does not require a generator-derived density.

Section 10.2 claims that path-space variational inference presupposes "a generator-derived density evaluable along paths." Relative entropy

$$D_{\mathrm{KL}}(P_{\mathrm{true},M} \| P_{\nu,M})$$

requires probability measures and suitable absolute continuity; it does not require either measure to arise from a generator. Non-Markovian and hidden-variable path laws can possess densities or Radon-Nikodym derivatives relative to suitable reference measures.

Consequently, failure of Chapman-Kolmogorov factorization does not establish that there is "nothing for a variational method to approximate." It establishes only that a Markov generator on the observed state space cannot reproduce the specified two-time kernels. The discussion should distinguish absence of a particular generator representation from absence of a path measure, likelihood, or variational target.

The two claimed "structural distinctions" in Table 2 do not survive in their present form.

Table 2 and its caption identify two distinctions:

- there is no generator-based true law to score against; and
- there is no generator-derived likelihood or posterior for a variational method to approximate.

Neither follows from indivisibility as stated. A true non-Markovian grid or path law may exist, as Remark 2.2 and Section 7 explicitly acknowledge. Likewise, a likelihood or posterior can be defined using a non-Markovian model, a hidden-state representation, or a deliberately misspecified surrogate.

What is established is narrower: no Markov generator on the observed state space reproduces the given indivisible two-time family. The entire comparison table should be revised around this precise statement.

The paper alternates between assuming only two-time kernels and assuming a full true path law.

The selection-functional column of Table 2 lists the assumed object as "two-time kernels $\Gamma(t \leftarrow s)$ only." However, Theorem 7.1 assumes a particular full grid law $P_{\mathrm{true},M}$. Since the two-time kernels do not uniquely determine this law, the framework must distinguish two different settings:

- only the two-time transition family is available, in which case general path observables under the true law are not identified; and
- an inducing process and its full grid law are specified, in which case path-space comparisons and KL divergences may become meaningful.

The table and concluding claims currently mix these settings.

The PAC-Bayes transfer is asserted without the required construction.

Section 10.3 claims that PAC-Bayes risk bounds "transfer to the calibrated surrogate level." However, the paper defines a point estimator

$$\widehat{\nu} = \arg\min_{\nu} \frac{1}{n} \sum_{i} S_{\nu,M}\left[\xi_{\mathrm{obs}}^{(i)}\right],$$

not a Gibbs posterior over ν. No prior on the parameter space, posterior distribution over ν, empirical-risk setting, or PAC-Bayes bound is formulated.

The similarity between $e^{-\lambda S}\Pi_M$ and a Gibbs measure on paths does not automatically supply a PAC-Bayes generalization theorem for $\widehat{\nu}$. This claim should either be removed or replaced with a precise theorem under the necessary sampling, prior, and loss assumptions.

The proposed multilevel extension conflicts with the claim that the grid is the model.

The multilevel-selection problem proposes coupling estimators across grids

$$M_0 < M_1 < \cdots$$

using multilevel Monte Carlo. Standard multilevel methodology requires estimators on successive levels to approximate a common limiting quantity, with controlled bias between levels. The manuscript repeatedly insists that the finite grid is the primary model and that no $M \to \infty$ limit has been established.

Until a renormalized continuum limit or another common target is defined, the selected measures and their estimands may simply change with M. There is then no established telescoping identity or level-bias decay on which a multilevel method could be based. This open problem should be made conditional on first solving the continuous-path or cross-grid consistency problem.

The grid-refinement calculations are correct, but their interpretation should be qualified.

The reported values

$$1 + \chi^2 \approx 37.7,\ 1373,\ 1.82 \times 10^6,\ 3.22 \times 10^{12}$$

for $M = 25, 50, 100, 200$, together with the corresponding values of Z_ν, follow from the stated determinant formulas. The approximate exponential rate $0.144M$ is also correct.

This convincingly demonstrates increasing mismatch between the particular proposal and Gibbs-selected measure under the chosen fixed-λ refinement. It is consistent with Proposition 2.3, but it should not be presented as a general grid-independent rate of continuum degeneracy. The rate depends on the proposal, surrogate, temperature, and discretization.

The empirical simulation does not independently verify the asymptotic theorems.

Figure 1 is consistent with the exact asymptotic variance, but 50 replications give a relatively noisy estimate of bias and variance, especially in the acknowledged heavy-weight-tail regime. The non-monotone estimated bias and the persistent underestimate of the asymptotic variance illustrate this limitation.

The figure is a useful numerical illustration, but phrases such as "bias decay illustrates Theorem 4.1(i)" should be softened: strong consistency is a pathwise asymptotic result and cannot be verified by a short finite-replication experiment.

The numerical experiment is not fully reproducible from the manuscript alone.

The closed-form values are reproducible from the formulas. However, the precise entries in Table 1, the plug-in estimate $\widehat{\chi}^2 \approx 623$, and the plotted empirical curves depend on the random-number generator, seed, and implementation details. None is supplied.

The data-and-code statement should distinguish reproducibility of the analytic benchmark from exact reproducibility of the Monte Carlo results. Supplying code and a fixed random seed would resolve this easily.

The "full risk trade-off" is not correctly stated.

Section 11 proposes the risk expression

$$\eta_{\nu,M}^2 + \frac{\sigma^2(\nu)}{N}.$$

From Theorem 7.1, the squared selection bias is bounded by

$$\big(F^*(\nu) - F_{\text{true}}\big)^2 \leq 4F_{\max}^2 \eta_{\nu,M}^2.$$

Moreover, the finite-N SNIS estimator generally has its own bias, so the exact mean-squared error relative to F_{true} contains additional terms. The displayed expression is therefore, at best, an asymptotic schematic trade-off. It should not be called the "full risk" without a derivation and the missing constants and terms.

The "optimal surrogate for an observable class" requires an objective over that class.

The asymptotic variance $\sigma^2(\nu)$ depends on the particular observable F. Thus,

$$\nu^\dagger = \arg\min_\nu \sigma^2(\nu)$$

defines an optimum for a fixed observable, not for an observable class. For a class $\mathcal{F}$, the author would need to specify an objective such as

$$\arg\min_\nu \sup_{F \in \mathcal{F}} \sigma_F^2(\nu)$$

or an averaged criterion.

Adaptive-proposal consistency does not follow automatically.

The statement that consistency under adaptive proposals "extends along standard lines" requires conditions controlling adaptation, dependence, weight stability, and proposal support. Since even the fixed-proposal example exhibits severe weight degeneracy, this extension may be nontrivial. It is reasonable as an open direction, but the assertion of consistency should be replaced by a description of what would need to be proved.

The temperature schedule in the continuous-path open problem is incomplete.

The notation "$\lambda_M \downarrow$" does not specify the proposed limiting value. Presumably the author intends $\lambda_M \downarrow 0$, but this should be stated. More importantly, an appropriate schedule would have to be derived rather than selected solely from the empirical blow-up rate in one example.

The reference list contains no obvious formatting problem.

The references are extensive and broadly relevant. However, the precise sample-size claims attached to Chatterjee-Diaconis, Sanz-Alonso, and the practical ESS discussion should be checked against the exact hypotheses and estimator types in those sources.